

# JOSEPH D VIVIANO

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## EDUCATION

- 2026– **PhD, Computer Science (incoming)** Mila [↗](#), Université de Montréal, QC/CA.
- Supervised by [Dr. Pierre Luc Bacon](#) [↗](#)  
Topic: Sample Efficient RL for High Dimensional Search.  
Funded by the NSERC Canada Graduate Scholarship – Doctoral.
- 2018–2020 **MSc., Computer Science** Mila [↗](#), Université de Montréal, QC/CA.
- Internship Supervised by [Dr. Yoshua Bengio](#) [↗](#)  
'Methods for controlling and utilizing saliency maps in medical imaging'.
- 2011–2013 **MSc. with Distinction, Biology** York University, ON/CA.
- Thesis Supervised by [Dr. Keith Schneider](#) [↗](#)  
'Tremotopic mapping of the human thalamic reticular nucleus'.
- 2005–2009 **BSc. Hons., Psychology** Queen's University, ON/CA.

## PROFESSIONAL EXPERIENCE

- 2022–Now **ML Scientist** Mila Québec AI Institute, QC/CA.
- Lead developer of [torchgfn](#) [↗](#) GFlowNet library (312 stars, 56 forks, 20 contributors).
  - GFlowNet application collaborations: Antibody design ([Amgen](#) [↗](#)), distributed search scaling ([Intel](#) [↗](#)).
  - Lab engineering support for ML for Science projects.
  - Grant writing: Mila's compute infrastructure, AI for Metabolic Engineering with [Joint BioEnergy Institute](#) [↗](#).
- 2021–2022 **ML Research Scientist** Deep Genomics [↗](#), QC/CA.
- Self-supervised learning for biological sequence representations.
  - Methods for cell-type generalization.
  - Drug screening anomaly detection pipeline.
- 2020–2021 **Applied Research Scientist** Mila Québec AI Institute, QC/CA.
- Consult on, propose, and implement deep learning solutions for Mila's partners: Optimal portfolio allocation (finance) & digital forgery-detection (fraud).
- 2020 **Research Intern** Google, PA/US.
- Uncertainty (epistemic) estimation methods for the [search ads predicted click through rate team](#) [↗](#).
- 2019 **Research Intern** Imagia, QC/CA.
- Developed methods for combining clinical notes with medical images to improve classification performance and for localizing disease without explicit labels of where the disease is located in an image.

2014–2017      **Research Methods Specialist**      Centre for Addiction and Mental Health, ON/CA.

- Biomarkers for vulnerable [schizophrenia](#) and [Alzheimer's](#) patients.
- Designed, built, and managed compute infrastructure and data management platform.

## AWARDS

2026      **Canada Graduate Scholarships – Doctoral**      NSERC.

Canada's most competitive graduate scholarship. \$40,000/year for 3 years.

2026      **Gold Reviewer Award**      ICML [Position Paper Track](#).

Awarded to the top-rated reviewers of the track.

## COMMUNITY SERVICE

2023–Now      **Area Chair**      Meta Reviewing.

Paper Adjudication for [NeurIPS Datasets and Benchmarks Track](#).

2014–Now      **Reviewing**      Paper reviews for conferences and journals.

Reviews for [ICLR](#), [NeurIPS Pre-registration in ML Workshop](#), [Biological Psychiatry](#), [MIDL](#), [PNAS](#), & [PLOS ONE](#).

## SELECTED PUBLICATIONS

[Viviano JD\\*](#), Younis OG\*, Choi S\*, Schmidt V, Bengio Y, Lahlou S. torchgfn: A PyTorch GFlowNet library. JMLR, 2026.

Boussif O, Ezzine LN, [Viviano JD](#), Koziarski M, Jain M, Malkin N, Bengio E, Assouel R, Bengio Y. Action abstractions for amortized sampling. ICLR, 2025.

Bertin P, [Viviano JD](#), Tejada-Lapuerta A, Wang W, Bauer S, Theis FJ, Bengio Y. A scalable gene network model of regulatory dynamics in single cells. arXiv, 2025.

Delaunay P, Bouthillier X, Breuleux O, Ortiz-Gagné S, Bilaniuk O, Normandin F, Bergeron A, Carrez B, Alain G, Blanc S, Osterrath F, [Viviano JD](#), et al. Introducing Milabench: Benchmarking Accelerators for AI. arXiv, 2024.

Cohen JP, [Viviano JD](#), Bertin P, Morrison P, Torabian P, Guarrera M, Lungren MP, Chaudhari A, Brooks R, Hashir M, Bertrand H. TorchXRyVision: A library of chest X-ray datasets and models. Medical Imaging with Deep Learning (MIDL), 2022.

Cohen JP, Cao T, [Viviano JD](#), Huang CW, Fralick M, Ghassemi M, Mamdani M, Greiner R, Bengio Y. Problems in the deployment of machine-learned models in health care. Canadian Medical Association Journal, 193(35), E1391-E1394.

Luccioni A, [Viviano JD](#). What's in the Box? A Preliminary Analysis of Undesirable Content in the Common Crawl Corpus. The Joint Conference of the Association for Computational Linguistics and the International Joint Conference on Natural Language Processing (ACL-IJCNLP), 2021.

[Viviano JD](#), Simpson B, Dutil F, Bengio Y, Cohen JP. Saliency is a Possible Red Herring When Diagnosing Poor Generalization. International Conference on Learning Representations (ICLR), 2021.

[Viviano JD](#), Buchanan RW, Calarco N, Gold J, Foussias G, Bhagwat N, Stefanik L, Hawco C, Malhotra AK, Voineskos AN, for the SPINS group. 2018. Resting-state connectivity biomarkers of cognitive performance and social function in schizophrenia spectrum disorders and healthy controls. Biological Psychiatry. 84(9), 665-674.

Bhagwat N, [Viviano JD](#), Voineskos AN, Chakravarty MM. 2018. Modeling and prediction of clinical

symptom trajectories in Alzheimer's disease using longitudinal data. 2018. PLoS Computational Biology. 14(9), e1006376.

DeSimone K, **Viviano JD**, Schneider KA. 2015. Population receptive field estimation reveals new retinotopic maps in human subcortex. Journal of Neuroscience, 35(27):9836-47.

**Viviano JD**, Schneider KA. 2015. Interhemispheric interactions of the human thalamic reticular nucleus. Journal of Neuroscience, 35(5):2026-32.

## INVITED TALKS

2024                      **Keynote Speaker**                                              FACSS SciX.  
Can an AI Understand a Scientist?

2023                      **Invited Speaker**                                              Joint Genome Institute Annual User Meeting.  
**Viviano JD**. Considerations for Machine Learning-Aided Lab in the Loop Scientific Discovery.